STAC WARS

PS 3

EXOPLANET HUNTERS

Operation Dim Star: find the Rebellion a new home



TRANSMISSION RECEIVED

A long time ago, in a galaxy far, far away…

The Death Star is gone, but the Empire is not. Every Rebel base is one betrayal away from being found, and Princess Leia is running out of places to hide.

Rebel scouts have found one last hope: three distant stars, STAC-1, STAC-2 and STAC-3, that dim in a strange, steady rhythm, as if something keeps passing in front of them. R2-D2 suspects planets. But the Alliance needs numbers, not suspicions: how big, how far, and could anyone live there?

The catch: the data is faint, and Imperial patrols are closing in. So Leia has called in the Alliance's newest analysts. That's **you**.

> *"Somewhere out there is a world the Empire hasn't found. Find it, and the Rebellion has a home." — Princess Leia*

YOUR MISSION

TOTAL: 15 MARKS

Astronomers have identified three distant stars — STAC-1, STAC-2 and STAC-3 — as potential hosts of exoplanets.

A space telescope has been observing these stars and recording their relative brightness over time. When a planet passes in front of its host star, the star's observed brightness decreases slightly. This phenomenon is known as a transit.

Your team has been provided with observational data for the three stellar systems. Analyse the data to:

- detect possible exoplanets,
- determine their physical and orbital properties, and
- identify the most promising candidate for further astronomical study.

Show all relevant calculations and clearly state your assumptions.

A. DETECT THE EXOPLANETS

4 MARKS

> *"R2 says every flicker is a clue. Don't ignore a single dip."*

Analyse the brightness observations for STAC-1, STAC-2 and STAC-3.

For each star:

1. Determine whether the observations indicate the presence of a transiting exoplanet.
2. If a periodic transit is detected, determine the approximate orbital period of the planet.
3. Determine the transit depth, i.e. the fractional decrease in stellar brightness.
4. Briefly explain why a periodic decrease in stellar brightness can indicate the presence of an exoplanet.

Use:

$$\delta = (F_{normal} - F_{transit}) / F_{normal}$$

where:
δ = transit depth
Fnormal = brightness outside transit
Ftransit = brightness during transit

B. DETERMINE THE PLANET SIZE

4 MARKS

> *"Judge a planet by its size, you do not. But the numbers, you must."*

For every system where a planet is detected, estimate the planet's radius using the transit depth.

Use:

$$\delta \approx (R_s / R_p)^2$$

where:
δ = transit depth
Rp = radius of planet
Rs = radius of star

Then:

5. Calculate the radius of each detected planet in kilometres.
6. Express the radius in terms of Earth's radius.

7. Classify each planet approximately as:
   - Earth-sized
   - Super-Earth
   - Neptune-sized
   - Jupiter-sized

C. DETERMINE THE PLANETARY ORBITS

4 MARKS

> *"Too close and it burns. Too far and it freezes. Get the distance right."*

Using the orbital period obtained in Part A and the mass of the host star, calculate the approximate orbital distance of each detected planet.

Use the simplified form of Kepler's Third Law:

$$P^2 = a^3 / M$$

where:
P = orbital period in years
a = orbital distance in AU
M = mass of the star in solar masses

Rearrange as required.

For each detected planet:

8. Calculate its orbital distance in AU.
9. Convert the distance into kilometres.
10. Compare its orbital distance with Earth's distance from the Sun.

D. FIND THE MOST PROMISING WORLD

3 MARKS

> *"The Rebellion needs a home, not a guess. Choose carefully."*

Using your results from Parts A–C and the stellar properties provided:

11. Determine which detected planet is the most promising candidate for further habitability studies.
12. Support your choice using at least three pieces of evidence from your analysis.
13. Suggest one additional observation or measurement that could help determine whether the planet has an atmosphere or could potentially support life.

For this question, use the simplified habitable-zone relation:

$$d_{HZ} \approx \sqrt{(L / L\odot)}\ AU$$

where:
dHZ = approximate distance of the habitable zone from the star
L = luminosity of the star
L ⊙ = luminosity of the Sun

> Important: A planet being within the habitable zone does not necessarily mean that it is habitable.

OBSERVATIONAL DATA

1. Stellar Properties

Property	STAC-1	STAC-2	STAC-3
Mass	1.00 M⊙	0.80 M⊙	0.50 M⊙
Radius	1.00 R⊙	0.80 R⊙	0.50 R⊙
Luminosity	1.00 L⊙	0.40 L⊙	0.08 L⊙
Surface Temperature	5800 K	5000 K	3800 K

2. STAC-1 — Brightness Observations

Time (days)	Rel. brightness	Time (days)	Rel. brightness	Time (days)	Rel. brightness
0	1.0000	7	1.0000	14	1.0000
1	1.0000	8	0.9970	15	1.0000
2	1.0000	9	1.0000	16	0.9970
3	0.9970	10	1.0000	17	1.0000
4	1.0000	11	1.0000	18	1.0000
5	1.0000	12	0.9970	19	1.0000
6	1.0000	13	1.0000	20	0.9970

3. STAC-2 — Brightness Observations

Time (days)	Rel. brightness	Time (days)	Rel. brightness
0	1.0000	120	1.0000
20	0.9990	140	1.0000
40	1.0000	160	1.0000
60	1.0000	180	0.9990
80	1.0000	200	1.0000
100	0.9990		

Note: The transit may occur between recorded observations. Use the repeated observations and the available data to estimate the period.

4. STAC-3 — Brightness Observations

Time (days)	Rel. brightness	Time (days)	Rel. brightness	Time (days)	Rel. brightness
0	1.0000	70	1.0000	140	0.9993
10	1.0000	80	1.0000	150	1.0000
20	0.9990	90	1.0000	160	1.0000
30	1.0000	100	0.9991	170	0.9990
40	1.0000	110	1.0000	180	1.0000
50	1.0000	120	1.0000	190	1.0000
60	0.9992	130	1.0000	200	1.0000

USEFUL CONSTANTS

- Solar radius, $R_\odot$ = 696,000 km
- Earth radius, $R_\oplus$ = 6,371 km
- 1 AU = 149.6 million km
- 1 year = 365.25 days
- 1 $M_\odot$ = mass of the Sun
- 1 $L_\odot$ = luminosity of the Sun

SUBMISSION REQUIREMENTS

Submit your analysis report (should contain calculations) in .pdf and IEEE formats containing:

14. Detection and identification of possible exoplanets.
15. Calculations of planetary radii.
16. Calculations of orbital distances.
17. Comparison of the detected planets and identification of the most promising candidate.
18. Clear reasoning supporting your conclusions.

All calculations must be shown. Answers without supporting calculations or reasoning may receive reduced marks.

MAY THE CODE BE WITH YOU